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Reporting & Performance Management

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Reporting & Performance Management

Operational Excellence, Glencore Copper
06 June 2022, Version 1.0

To close out the drafting of the manual a workshop was held with the following stakeholders in May 2022:

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Supported by the AM Network Pty Ltd and Work Management Solutions Pty Ltd

Preface

The Asset Management Framework, Policy and Manuals are intended to provide leading Asset Management practices when applied and will be continually improved for the benefit of Glencore Copper.

The real value of sharing the knowledge gained from the use of these documents will come from continued discussion of the issues and rationale behind each document rather than by reading and interpreting them as individuals.

Readers are encouraged to consider these against our values around Safety, Responsibility, Simplicity, Integrity, Openness and Entrepreneurialism and how they reflect our Purpose – to responsibly source the commodities that advance everyday life. This document lays out the minimum standard for Glencore Copper and compliance to this manual will demonstrate a maturity value equal to Planned/Deployed (3) on the Glencore Copper Asset Management Maturity Scale.

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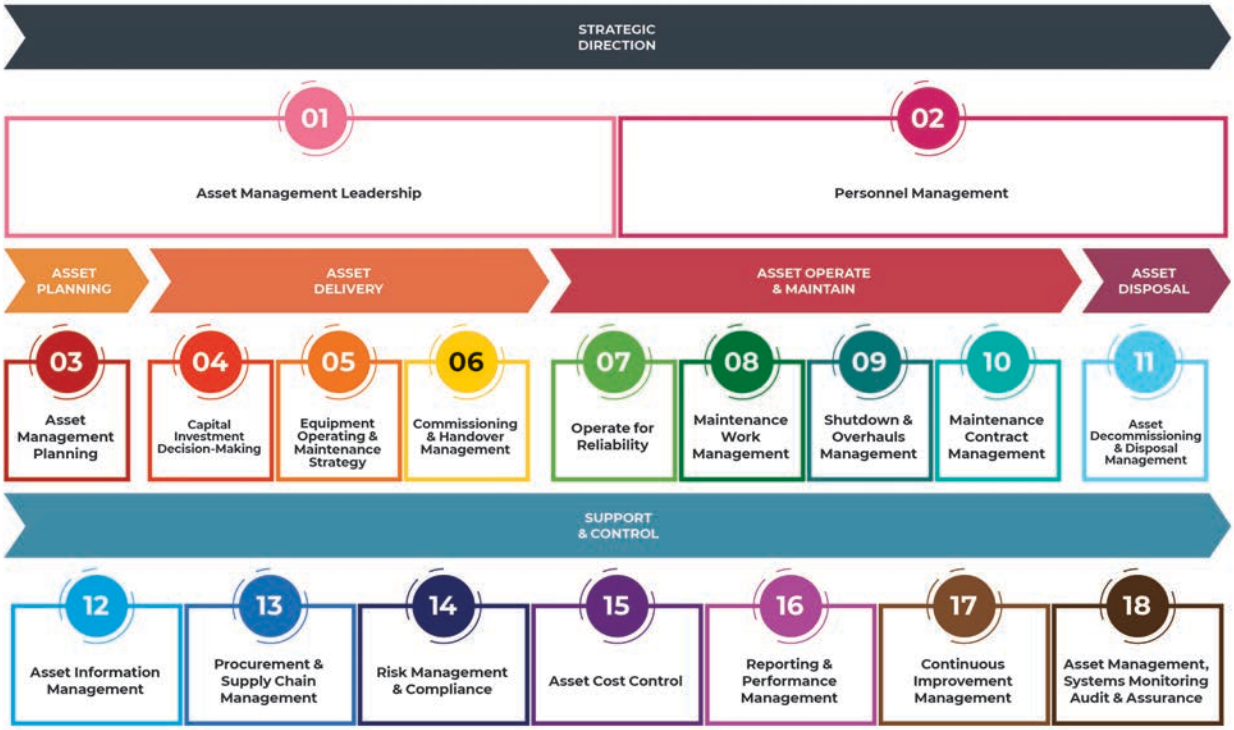
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1. Introduction

The Asset Management Policy and Manuals define how Glencore Copper manages the whole of life of an asset to ensure optimal value is extracted by ensuring assets are capable of meeting business and stakeholder objectives.

Figure 1-1 describes the high-level Asset Management Framework, which enables Glencore Copper to deliver its Asset Management purpose and vision.

FIGURE 1-1 ASSET MANAGEMENT FRAMEWORK



2. Objectives

The objective of this manual for Reporting and Performance Management is to define how Glencore Copper records, measures, analyses, reports on asset performance and asset management related metrics across all operating sites.

The drivers of the above objective include but are not limited to the following:

- Ability to make decisions (tactical or strategic) based on complete, accurate, consistent and current information that is readily accessible to enable personnel to perform their jobs effectively and help to create a professional, results-oriented culture.
- Up to date and accurate metrics and reports are readily available to the right level in the business to improve decision-making.
- Information captured is validated before analysis is carried out and reports are issued.
- Continuous improvement relies on consistency and availability of information.
- Timely and accurate information allows quick feedback on improvements and supports an effective work environment.
- Reporting on Global Metrics and closing of gaps to targets is a business requirement.

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- Reporting and performance management enables the exchange and replication of good practices between sites.
- Ensuring Metrics from each Element are implemented effectively.

2.1 Accountabilities

Dedicated resources must be assigned and capabilities developed within each Operation to deliver Reporting and Performance Management processes that enable effective delivery of the Asset Management objectives.

Assign single point accountability for each or all the following:

- Asset Management Metrics and Targets Aligned
- Asset Management Data Captured
- Asset Management Metrics Results
- Metric Variation Analysis
- Metric Variation Management
- Metrics Review

2.2 Scope

This manual is to be read in conjunction with all other manuals in the Asset Management Framework and the Glencore Copper Asset Management Metrics Standard.

The scope includes:

- Ensuring asset performance targets are aligned with the business plan objectives.
- Ensuring data captured for the Asset Management metrics is preferably available directly from a system(s) solution in lieu of manual data capture and data is complete, accurate, consistent, timely and accessible.
- Ensuring risks are identified and managed appropriately.
- Responsibility for reporting and performance management decision-making is clear.
- Ensure decisions are made at the right levels in the business to add value.
- Ensure decisions are timely and cost-effective.
- Performance measures and targets exist and are communicated to all stakeholders.
- Performance measures are consistent across all business units and enable benchmarking between sites.
- Operating context, asset condition and capability are considered in managing performance variations and actions to resolve any shortfalls (Planned vs Actual performance) are identified and implemented.
- Continuous Improvement opportunities are clearly identified and managed appropriately based on complete, accurate, consistent, timely and accessible information.

3. Definitions

Continuous Improvement (CI) – is an ongoing, long-term approach to improving processes, products and services.

Information – processed and organised data, which has some meaning or purpose and is typically processed with a particular aim or objective in mind, i.e. for decision making purposes.

Lagging Metric - is an observable or measurable factor that changes sometime after the asset management variable with which it is correlated changes.

Leading Metric - measure change and deal with immediate progress and show the likelihood that the goals will be achieved.

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Metrics – measures of quantitative assessment commonly used for assessing, comparing and tracking performance, risk and production.

Time Usage Model – defines the standard allocation of downtime and rate loss and forms the basis for time-based asset performance metric calculations.

Performance Measurement - the process of collecting, analysing and/or reporting information regarding the performance of an individual, group, organization, system or component.

Stakeholder – is either an individual, group or organisation who is impacted by the outcome of a project, activity or decision.

4. Approach

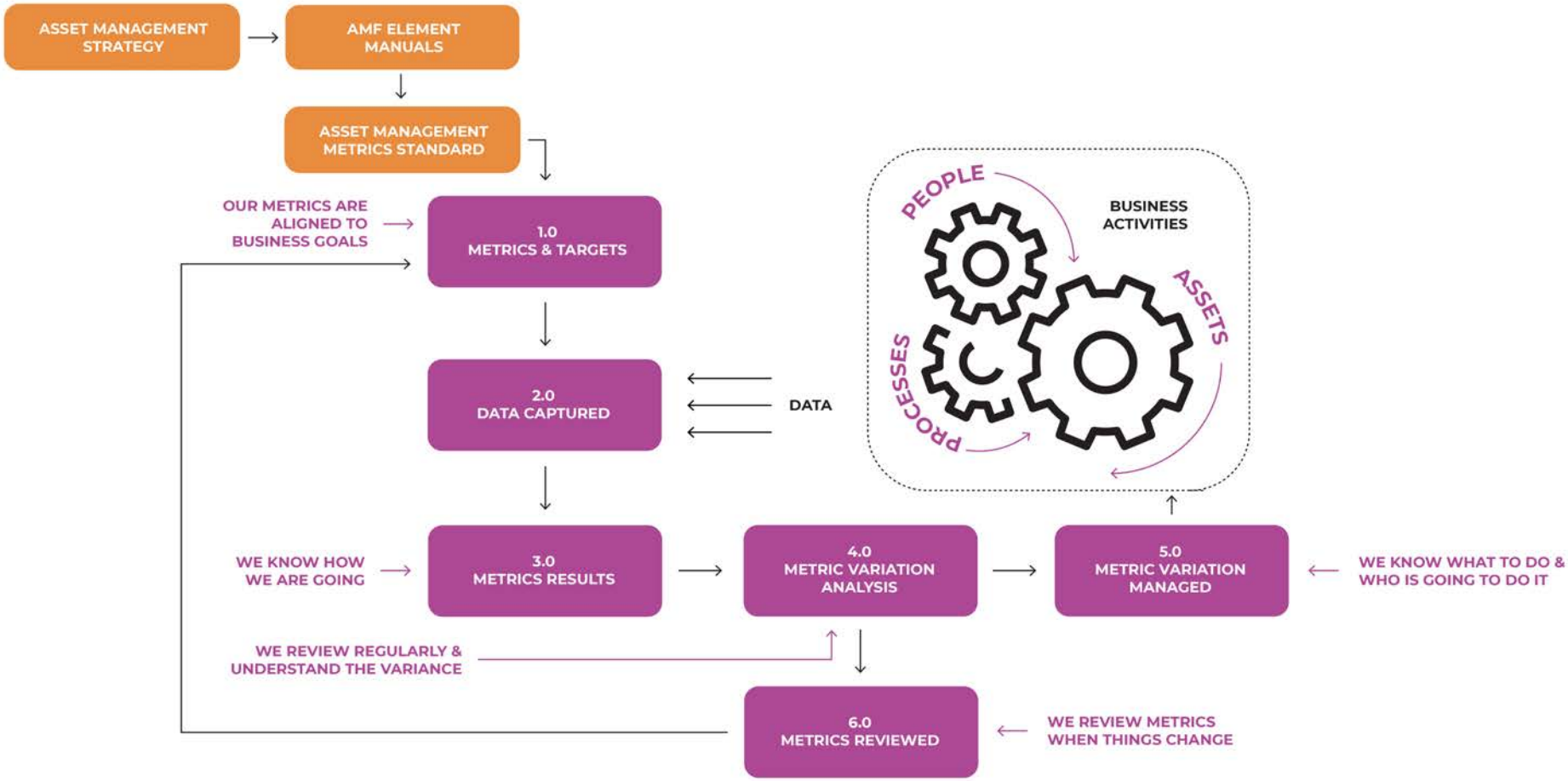
The approach for Reporting and Performance Management is described through:

- Process Flow - pictorially illustrates the key Reporting and Performance Management activities, refer to Figure 4-1.
- Process Description – defines in more detail the key Criticality Analysis activities, refer to Section 4.2.
- Process RACI Matrix - indicates for each activity, the Functions who are responsible, Accountable, who are Consulted prior to final decision/action and those who are Informed of any outcomes and actions, refer to Section 4.3.

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4.1 Process Flow

FIGURE 4-1 REPORTING AND PERFORMANCE MANAGEMENT PROCESS



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4.2 Process Description

TABLE 4-1 REPORTING AND PERFORMANCE MANAGEMENT PROCESS DESCRIPTION

PROCESS STEP	DESCRIPTION
1.0 Metrics and Targets	**Establishing and implementing key measures and targets in line with the Glencore Copper Asset Management Metrics Standard.**
1.1 Site Measures/Targets Established	The Glencore Copper Asset Management Framework consists of a Framework document and 18 supporting manuals (one per Element) as well as an Asset Management Metrics Standard that together provides Glencore Copper with a suite of corporate level metrics for implementation. Consideration should be given to both leading and lagging metrics. In addition to the corporate metrics, each site may identify and implement metrics to realise site specific opportunities aligned to asset management objectives. When developing measures and targets the following should be considered: • Showing the alignment of measures with asset management objectives/strategic goals • Reflecting a balance among competing demands • Presenting only the information that the audience requires • Keeping the number of measures manageable • Updating and maintaining relevance as maturity grows. Metrics guidance material shall be available to articulate: • What is being measured – Definition • Why it is being measured – Rationale • Target – Performance Indicator • How it is calculated – Calculation • When it is calculated – Time basis. Refer to Appendix A for the list of corporate metrics derived from the suite of Glencore Copper Manuals.
1.2 Site Aligned to Asset Management Metrics	Metrics and targets shall be communicated and cascaded throughout each site. This provides a line of sight for each team and individual to the site metrics.
1.3 Full Asset Potential Defined	Today's business environment demands that each site embraces operational discipline and understands their business drivers to remain competitive in changing market conditions. Asset management aligns with these principles by seeking to create value to shareholders by optimising reliability, eliminating losses, maximising utilisation and asset life in a safe, cost-effective manner. It does this by striving to move from a reactive to proactive work environment where effective decisions (either tactical or strategic) are made throughout the life cycle of an asset from business need, through acquisition or modification, into operation and maintenance and finally to renewal and/or disposal. . This is achieved through: • People – determine what capabilities and competencies that are needed to grow • Processes – determine what processes and practises are needed to excel • Assets – asset performance requirements to meet production demands • Finance – how do shareholders value Glencore Copper • Business – what do Glencore Copper aspire to. The Asset Management Maturity Self-Assessment (AMMSA) Guideline and Tool shall be used annually as part of the business planning cycle to assess maturity and to identify and

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	implement improvement opportunities, in line with E17 Continuous Improvement Management. Continuously improving Asset Management Practices will increase Glencore Copper's confidence to achieve full asset potential/value. This is summarised in the graphic below: 
2.0 Data Captured	**Capturing and consolidating asset data through multiple systems (with each metric having one source of the truth)**
2.1 Accurate Consistent Asset Management Metrics	Asset information shall be validated to ensure completeness, accuracy and consistency to enable up to date metrics and reports to the right level in the business to improve decision-making.
2.2 Multiple Systems Considered	Using multiple systems to capture and consolidate asset management information, such as: • CMMS • Process and Production Systems • Online Asset Health Monitoring • Asset Performance Management System. Each metric developed shall have a single source of the truth and be readily accessible regardless of what system information is used to report on the metric. Manual data capture is non preferred and where this may be the case, system solutions shall be investigated and implemented where appropriate.
3.0 Metrics Results	**Metric results determine how the business is performing**
3.1 Determine the Metric Results	Asset Management metrics results are to be established and correlated into a useable format for communicating performance and to allow variation analysis through an automated process.
4.0 Metric Variation Analysis	**Assigning responsibility and developing competencies for variation management of Asset Management Metrics.**
4.1 Responsibility Assigned to Variance	Performance management responsibilities shall be assigned for reviewing and reporting on metric variance and for assigning corrective actions against each Asset Management Metric variance. Competence shall be assessed for each person responsible for metric variance control and action plans put in place to upskill responsible people if required. Metrics guidance material should be developed on how to influence improvement to variations.
4.2 Review Cycle Established	Asset Management performance review cycles are to be established with a schedule of meetings, agendas, required attendees, required information inputs and a managed process for actions.

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	The review should consider both variance from the required level of performance and also reporting, communicating and celebrating success.
4.3 Reporting Tools Implemented	Each site is responsible for the establishment of asset management performance reporting tool/s, data analysis methods and ensure a program is in place to establish competencies to manage the asset management performance data and reporting requirements.
4.4 Variation Analysis	Correlated metrics are to be analysed and variations investigated and understood. The site shall determine the method for variation analysis.
5.0 Metric Variation Managed	**Reporting and sharing Asset Management Metrics Information to stakeholders to enable variation to be managed.**
5.1 Reporting Protocols Established	Reporting protocols are to be established that should include: • Format of each report • Who is responsible for generating and distributing the reports • When reports are disseminated • Who receives the reports • How the metrics information is used to drive improvements • Who is responsible for improving each report • Who is responsible for managing each variance.
5.2 Metrics Information Shared	Metrics information shall be shared using the following means of communication: • Notice boards, newsletters and communication meetings • Start of shift meetings and toolbox talks • Operations planning meetings (Daily, weekly, medium term) • Maintenance and Engineering performance reviews • Budget reviews • Shutdown reviews. Communities of practices should also be promoted to share metrics information across sites globally to accelerate asset management improvements.
6.0 Metrics Reviewed	**Developing and implementing a system to manage reviews and improvements to Asset Management Metrics.**
6.1 Manage Asset Management Metrics Improvement	A system for managing Asset Management Metrics improvement shall be established considering: • Changes to site business objectives and/or asset management objectives • Changes to site business processes • Technology advances • Asset Management Maturity • Lessons learned
6.2 Review Metrics	Conduct a review of all the asset management metrics, at intervals to be established at each site to ensure that the metrics are still valid and fit for purpose.

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4.3 Process RACI Description

The RACI matrix below in Table 4-2 addresses the high-level functional elements of the Reporting and Performance Management process. For the overall Asset Management RACI matrix, refer to the Asset Management Intranet page.

Note: The HSE and Commercial Departments do not appear in this high-level RACI, although their involvement will be required from time to time dependent on the nature of the variance.

The codes used in the RACI analysis are defined as follows:

TABLE 4-2 PROCESS RACI

R	Responsible	The "doer" – the individual(s) who will perform an activity or directs others to perform the activity.
A	Accountable	"The buck stops here" – the individual who ultimately "owns" the activity – there should only be one person
C	Consulted	Person that needs to provide feedback and contribute to the activity
I	Inform	The individual(s) who needs to be informed after a decision or action is taken if relevant. (via 1-way communication)

Process Step \ Function	General Management	Maintenance Management	Operations Management	Maintenance Execution	Supply / Warehouse	Maintenance Planning	Technical Services	Reliability
1.0 Metrics and Targets								
1.1 Site Measures/Targets Established	C	A	C	R	R	C	R	R
1.2 Site Aligned to Asset Management Metrics	C	A	C	R	C	C	R	R
1.3 Full Asset Potential Defined	C	A	R	R	R	C	R	R
2.0 Data Captured								
2.1 Accurate Consistent Asset Management Metrics	C	A	R	R	R	R	R	R
2.2 Multiple Systems Considered	C	A	R	R	R	R	R	R
3.0 Metrics Results								
3.1 Determine the Metric Results	I	A	R	R	C	C	R	R
4.0 Metrics Variation Analysis								
4.1 Responsibility Assigned to Variance	I	A	C	R	C	C	R	R
4.2 Review Cycle Established	I	A	C	R	C	C	R	R
4.3 Reporting Tools Implemented	C	A	C	R	C	C	R	R
4.4 Variation Analysis	C	A	C	R	C	C	R	R
5.0 Metric Variation Managed								
5.1 Reporting Protocols Established	I	A	C	R	C	C	R	R
5.2 Metrics Information Shared	I	A	I	R	C	C	R	R
6.0 Metrics Reviewed								

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Process Step \ Function	General Management	Maintenance Management	Operations Management	Maintenance Execution	Supply / Warehouse	Maintenance Planning	Technical Services	Reliability
6.1 Manage Asset Management Metrics Improvement	C	A	C	R	C	C	R	R
6.2 Review Metrics	I	A	C	R	C	C	R	R

TABLE 4-3 FUNCTIONAL DESCRIPTIONS

Function	Description
General Management	• Provides input regarding the alignment of Asset Management metrics to the Glencore Copper Asset Management Metrics Standard. • Provides input regarding the accuracy of Asset Management metrics and the systemisation of the metrics. • Provides input into the management of Asset Management metrics regarding variance, reviewing, reporting and information distribution. • Responsible with respect to the review and improvement of Asset Management metrics. • Provides input into the reporting of Asset Management metrics regarding protocols, responsibilities, procedures and stakeholder input. • Responsible with respect to establishing reporting protocols. • Responsible with respect to analysing metrics variation.
Maintenance Management	• Ensures that alignment of Asset Management metrics to the Glencore Copper Asset Management Metrics Standard regarding the establishment of metrics, other asset management systems metrics, that the full potential of assets are defined and assets are managed accordingly to reach that potential. • Ensures that asset information is captured regarding the accuracy and consistency of metrics and that metrics are systemised. • Ensures that variations in the performance of asset and processes through metrics are managed regarding assigning responsibility for the variance, establishing review cycles, implementing reporting tools and the process for the distribution of information regarding metrics. • Ensures that asset management metric results are determined. • Ensures improvement of Asset Management metrics. • Ensures reporting protocols, responsibilities, procedures are established for Asset Management metrics and stakeholder input are considered. • Ensures that asset management metrics are reviewed.
Maintenance Execution	• Provides input regarding the alignment of Asset Management metrics to the Glencore Copper Asset Management Metrics Standard and is responsible for established other asset management system metrics and that that the full potential of assets is defined and assets are managed accordingly to reach that potential. • Responsible for the capture of asset information, regarding its accuracy and consistency of metrics and that metrics are systemised. • Responsible with respect to the management of Asset Management metrics regarding variance and review cycles and is responsible for establishing reporting tools and a process for information distribution. • Responsible with respect to determining asset management metrics results. • Responsible for improvement of asset management metrics. • Responsible the establishment of reporting protocols, responsibilities, procedures for Asset Management metrics and stakeholder input are considered.

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Maintenance Planning	• Provides input regarding the alignment of Asset Management metrics to the Glencore Copper Asset Management Metrics Standard. • Provides input regarding defining full potential of assets and assets are managed accordingly to reach that potential. • Responsible for captured of asset data, regarding the accuracy and consistency of metrics and that metrics are systemised. • Provides input into the management of Asset Management metrics regarding variance, reviewing, reporting and information distribution. • Provides input regarding improvement of Asset Management metrics. • Provides input into the establishment of reporting protocols, responsibilities, procedures for Asset Management metrics and stakeholder input are considered.
Operations Management	• Provides input with regards to the alignment of Asset Management metrics to the Glencore Copper Asset Management Metrics Standard regarding the establishment of metrics, other asset management systems metrics, that the full potential of assets are defined and assets are managed accordingly to reach that potential. • Responsible with respect to defining asset full potential. • Responsible with respect to determine asset management metrics results. • Responsible for the capture of asset information regarding the accuracy and consistency of metrics and that metrics are systemised. • Provides input into the management of Asset Management metrics regarding reviewing, reporting and information distribution. • Provides input to the management of variances and improvement of asset management metrics. • Provides input into the establishment of reporting protocols, responsibilities, procedures for Asset Management metrics and stakeholder input are considered.
Supply/Warehouse	• Provide input to the alignment of Asset Management metrics to the Glencore Copper Asset Management Metrics Standard and established other asset management system metrics and that that the full potential of assets are defined and assets are managed accordingly to reach that potential. • Responsible for Time Usage Model alignment. • Responsible for the capture of asset information, regarding its accuracy and consistency of metrics and that metrics are systemised. • Provides input to the management of variances and improvement of asset management metrics. • Provides input regarding determine asset management metrics results. • Provides input into the establishment of reporting protocols, responsibilities, procedures for Asset Management metrics and stakeholder input are considered.
Reliability	• Responsible for the alignment of Asset Management metrics to the Glencore Copper Asset Management Metrics Standard and established other asset management system metrics and that that the full potential of assets are defined and assets are managed accordingly to reach that potential. • Responsible for Time Usage Model alignment. • Responsible for the capture of asset information, regarding its accuracy and consistency of metrics and that metrics are systemised. • Responsible for improvement of asset management metrics. • Responsible for assigning responsibilities for variances regarding metric performance. • Responsible with respect to determining asset management metrics results. • Responsible with respect to the establishment of reporting protocols, responsibilities, procedures for Asset Management metrics and stakeholder input are considered. • Responsible with respect to the review of asset management metrics.
Technical Services -Mining, Processing, Engineering	• Responsible for the alignment of Asset Management metrics to the Glencore Copper Asset Management Metrics Standard and established other asset management system metrics and

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that that the full potential of assets are defined and assets are managed accordingly to reach that potential.
- Responsible for Time Usage Model alignment.
- Responsible for the capture of asset information, regarding its accuracy and consistency of metrics and that metrics are systemised.
- Responsible for improvement of asset management metrics.
- Responsible for assigning responsibilities for variances regarding metric performance.
- Responsible with respect to determining asset management metrics results.
- Responsible with respect to the establishment of reporting protocols, responsibilities, procedures for Asset Management metrics and stakeholder input are considered.
- Responsible with respect to the review of asset management metrics.

5. Deployment

The deployment of the Reporting and Performance Management approach across all business areas is through:

- Awareness presentation
- Position description alignment
- Competency assessment
- Training, coaching and mentoring

5.1 Awareness Presentation

This Manual is supported by a slide presentation and video to help facilitate communication and to raise awareness on how it contributes to the Asset Management Framework and business success.

5.2 Position Description Alignment

To effectively deploy this Manual, the essential process steps and activities need to be diligently mapped and assigned to approved positions within Glencore Copper.

The RACI matrix assists in the visualisation that everyone is working towards the same goal. It clarifies, not only the individual's responsibilities, but also that of others. It has proven to be an effective tool to quickly understand and close the gaps between current and desired practices and business process activities.

Position descriptions including metrics need to be aligned to the RACI matrix and measures of success. This is the basis for ongoing performance reviews and the creation and execution of individual development plans.

5.3 Competency Assessment

This Manual is supported by an Asset Management competencies assessment tool which is used to:

- Guide recruitment, selection, development and deployment of people in asset management roles across the company.
- Provide employees responsible for managing assets with feedback, development, coaching and recognition for their performance.
- Map, build and manage the asset management talent pool across the company.
- Identify strengths and development needs in asset management practitioners.

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5.4 Training, Coaching and Mentoring

Coaching and mentoring are another way to assist employees improve their understanding of their role and to improve performance.

Coaching is generally about improving performance in the short term where mentoring is about achieving high performance and career aspirations over a longer term.

Coaching or mentoring requires addressing that a person is not currently up to the required level to perform their role as effectively and efficiently as they could. The build-up of skills to perform efficiently and effectively requires a large investment of time and cooperation.

Coaches and Mentors need to be carefully selected and must be committed for the long term to effect maximum success.

6. Improvement

The process to review and continually improve the approach and deployment for Reporting and Performance Management is scheduled Annually, or when there is a sudden change in the Operating Context. As the organization matures a longer review period could be considered.

7. References

The following documents provide detailed support material that should be read in conjunction with this Manual to provide a comprehensive understanding of Reporting and Performance Management.

TABLE 7-1 REFERENCES

DOCUMENT TITLE	DESCRIPTION
Asset Management Framework	This document defines the overall direction of asset management across Glencore Copper.
Site AMIP Integration Approach	Presentation outlining the Asset Management Improvement Plan Integration approach.
Asset Management Metrics Standard	This document defines the suite of Asset Management Metrics for Glencore Copper and the Time Usage Model.

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8. Revision History

VERSION NO.	REASON FOR REVISION	DATE
0.1	New Manual to support the Asset Management Framework	30/07/2021
0.2	Reformat and Issue for Final Review	18/05/2022
1.0	Update Site Signoff and Publish	06/06/2022

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9. Appendix A – Asset Management Corporate Metrics



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10. Appendix B – Stakeholder Signoff

Stakeholder Name	Site or Department	Contact	Stakeholder Signoff
Anthony Green	Asset Management - Australia	Anthony.green@glencore.com.au	9/7/2022
Akeem Araoye	Asset Management – North America	Akeem.Araoye@glencore.ca	9/7/2022
Yuri Yurac	Asset Management – South America	Yuri.yurac@glencore.cl	8/7/2022
Lawrence Hlatshwayo	Asset Management – Africa	Lawrence.hlatshwayo@glencore.co.za	8/7/2022
Rohan Reisener	Asset Management - Australia	Rohan.reisener@glencore.com.au	7/7/2022
Dino Moll	Asset Management - Australia	Dino.Moll@glencore.com.au	10/7/2022

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Reporting & Performance Management

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